SIMATS ENGINEERING
Engineer to Excel

SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Course: Fundamentals of Data Science

Course Code: DSA0404

Slot: A

Faculty: KRISHNARAJ M

Submitted By

S.Siva Charan (192472147)

Lab Programs

1.student_average.py

Code:

student_average.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDS lab/student_average.py (3.12.3)

File Edit Format Run Options Window Help

```
import numpy as np

student_scores = np.loadtxt(
    "Students_data.csv",
    delimiter=",",
    skiprows=1,
    usecols=(1, 2, 3, 4)
)

subjects = ["Math", "Science", "English", "History"]

average_scores = np.mean(student_scores, axis=0)

highest_subject = subjects[np.argmax(average_scores)]

print("Average Scores:", np.round(average_scores, 2))
print("Subject with Highest Average Score:", highest_subject)
```

Output:

IDLE Shell 3.12.3

File Edit Shell Debug Options Window Help

```
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDS lab/student_average.py
Average Scores: [82.  88.  83.2 84.8]
Subject with Highest Average Score: Science
>>>
```

2.Past_month_price_average.py

Code:

past_month_price_average.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDS lab/past_month_price_average.py (3.12.3)

File Edit Format Run Options Window Help

```python
import pandas as pd

df = pd.read_csv("Saless_data.csv")

average_price = df[df["Month_sales"] == "November 2023"]["Price"].mean()

print("Average Price in November 2023:", round(average_price, 2))
```

Output:

IDLE Shell 3.12.3

File Edit Shell Debug Options Window Help

```
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> 
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDS lab/past_month_price_average.py
Average Price in November 2023: 25000.0
>>> 
```

3. house_average_price.py

house_average.py - C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\FDS lab\house_average.py (3.12.3)

File Edit Format Run Options Window Help

```python
import numpy as np

house_data = np.loadtxt(
    "House_data.csv",
    delimiter=",",
    skiprows=1
)

houses_more_than_4 = house_data[house_data[:, 1] > 4]

average_price = np.mean(houses_more_than_4[:, 3])

print("Average Price of Houses with more than 4 Bedrooms:", round(average_price, 2))
```

Output:

IDLE Shell 3.12.3

File Edit Shell Debug Options Window Help

```
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> 
= RESTART: C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\FDS lab\house_average.py
Average Price of Houses with more than 4 Bedrooms: 15500000.0
>>> 
```

4.quarterly_sales_numpy.py

Code:

```
quarterly_sales_numpy.py - C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\FDS lab\quarterly_sales_numpy.py (3.12.3)
File  Edit  Format  Run  Options  Window  Help
import numpy as np

sales_data = np.genfromtxt(
    "Quarterly_sales_data.csv",
    delimiter=",",
    skip_header=1,
    dtype=str
)

months = sales_data[:, 1]
sales = sales_data[:, 4].astype(float)

Q1 = 0.0
Q4 = 0.0

for i in range(len(months)):
    if months[i] in ["January", "February", "March"]:
        Q1 += sales[i]
    elif months[i] in ["October", "November", "December"]:
        Q4 += sales[i]

total_sales_year = np.sum(sales)

percentage_increase = ((Q4 - Q1) / Q1) * 100

print("Total Sales for the Year:", round(total_sales_year, 2))
print("Percentage Increase from Q1 to Q4:", round(percentage_increase, 2), "%")
```

Output:

```
IDLE Shell 3.12.3
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\FDS lab\quarterly_sales_numpy.py
Total Sales for the Year: 590000.0
Percentage Increase from Q1 to Q4: 50.0 %
>>>
```

5. fuel_efficiency_analysis.py

Code:

```
fuel_efficiency_analysis.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDS lab/fuel_efficiency_analysis.py (3.12.3)
File  Edit  Format  Run  Options  Window  Help
import numpy as np

fuel_data = np.genfromtxt(
    "Fuel_data.csv",
    delimiter=",",
    skip_header=1,
    dtype=str
)

fuel_efficiency = fuel_data[:, 7].astype(float)
make = fuel_data[:, 8]

average_efficiency = np.mean(fuel_efficiency)

mazda_eff = fuel_efficiency[make == "mazda"]
audi_eff = fuel_efficiency[make == "audi"]

mazda_avg = np.mean(mazda_eff)
audi_avg = np.mean(audi_eff)

percentage_improvement = ((mazda_avg - audi_avg) / audi_avg) * 100

print("Average Fuel Efficiency of all cars:", round(average_efficiency, 2), "MPG")
print("Average Fuel Efficiency of Mazda:", round(mazda_avg, 2), "MPG")
print("Average Fuel Efficiency of Audi:", round(audi_avg, 2), "MPG")
print("Percentage Improvement from Mazda to Audi:", round(percentage_improvement, 2), "%")
```

Output:

```
IDLE Shell 3.12.3
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDS lab/fuel_efficiency_analysis.py
Average Fuel Efficiency of all cars: 28.67 MPG
Average Fuel Efficiency of Mazda: 31.0 MPG
Average Fuel Efficiency of Audi: 27.0 MPG
Percentage Improvement from Mazda to Audi: 14.81 %
>>>
```

6. customer_purchase.py

Code:

```
customer_purchase.py - C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\FDS lab\customer_purchase.py (3.12.3)
File  Edit  Format  Run  Options  Window  Help
import pandas as pd

df = pd.read_csv("grocerystore.csv")

total_sales = df["Sales"].sum()

print("Total sales for the grocery store:", round(total_sales, 2))

discount = 0.10
tax = 0.18

price_after_discount = total_sales - (total_sales * discount)

price_after_tax = price_after_discount + (price_after_discount * tax)

print("Final amount after applying discount and tax:", round(price_after_tax, 2))
```

Output:

```
IDLE Shell 3.12.3
File  Edit  Shell  Debug  Options  Window  Help
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\FDS lab\customer_purchase.py
Total sales for the grocery store: 3850
Final amount after applying discount and tax: 4088.7
>>>
```

7.customer_order_analysis.py

Code:

customer_order_analysis.py - C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\FDS lab\customer_order_analysis.py (3.12.3)

File Edit Format Run Options Window Help

```
import pandas as pd

df = pd.read_csv("order_data.csv")

total_orders = df.groupby("Full Name")["Order Count"].sum()

avg_order_per_product = df.groupby("Items")["Order Total"].mean()

earliest_order = df["Order Date"].min()

latest_order = df["Order Date"].max()

print("Total number of orders made by each customer:")
print(total_orders)

print("\nAverage order total for each product:")
print(avg_order_per_product)

print("\nEarliest order date:", earliest_order)

print("Latest order date:", latest_order)
```

Output:

IDLE Shell 3.12.3

File Edit Shell Debug Options Window Help

```
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\FDS lab\customer_order_analysis.py
Total number of orders made by each customer:
Full Name
Anjali    3
Priya     3
Rahul     5
Name: Order Count, dtype: int64

Average order total for each product:
Items
Headphones     4000.0
Laptop        75000.0
Mobile        40000.0
Name: Order Total, dtype: float64

Earliest order date: 2023-01-10
Latest order date: 2023-05-18
>>>
```

8. five_most_sold_products.py

Code:

five_most_sold_products.py - C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\FDS lab\five_most_sold_products.py (3.12.3)

File Edit Format Run Options Window Help

```
import pandas as pd

df = pd.read_csv("Electronics_data.csv")

top_5_products = df.groupby("Product_Name")["Price"].sum().nlargest(5)

print("Five Most Sold Products:")
print(top_5_products)
```

Output:

IDLE Shell 3.12.3

File Edit Shell Debug Options Window Help

```
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\Users\Sravan Kumar.Y\AppData\Local\Programs\Python\Python312\FDS lab\five_most_sold_products.py
Five Most Sold Products:
Product_Name
Laptop          165000
TV               62000
Mobile           45000
Refrigerator     45000
AC               40000
Name: Price, dtype: int64
>>>
```

Course: Fundamentals of Data Science

Course Code: DSA0404

Slot: A

Faculty: KRISHNARAJ M

Submitted By

S.Siva Charan (192472147)

Lab Programs

9.Real Estate Property

CODE

RealEstate.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/RealEstate.py (3.12.3)

File Edit Format Run Options Window Help

```
import pandas as pd

# Input data
property_data = pd.DataFrame({
    'Property_ID': [101, 102, 103, 104, 105],
    'Location': ['Chennai', 'Bangalore', 'Chennai', 'Hyderabad', 'Bangalore'],
    'Bedrooms': [3, 5, 4, 6, 2],
    'Area_sqft': [1200, 2500, 1800, 3000, 1000],
    'Listing_Price': [6000000, 12000000, 8500000, 15000000, 5000000]
})

# 1. Average listing price in each location
avg_price = property_data.groupby('Location')['Listing_Price'].mean()

# 2. Number of properties with more than four bedrooms
count_properties = (property_data['Bedrooms'] > 4).sum()

# 3. Property with the largest area
largest_property = property_data.loc[property_data['Area_sqft'].idxmax()]

print("1. Average Listing Price by Location:")
print(avg_price)

print("\n2. Number of Properties with More Than Four Bedrooms:")
print(count_properties)

print("\n3. Property with the Largest Area:")
print(largest_property)
```

OUTPUT

IDLE Shell 3.12.3

File Edit Shell Debug Options Window Help

```
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/RealEstate.py
1. Average Listing Price by Location:
Location
Bangalore      8500000.0
Chennai        7250000.0
Hyderabad     15000000.0
Name: Listing_Price, dtype: float64

2. Number of Properties with More Than Four Bedrooms:
2

3. Property with the Largest Area:
Property_ID             104
Location          Hyderabad
Bedrooms                  6
Area_sqft              3000
Listing_Price      15000000
Name: 3, dtype: object
>>>
```

Experiment 10

1.Monthly Sales Visualization

CODE

Monthlysales.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/Monthlysales.py (3.12.3)

File Edit Format Run Options Window Help

```python
import matplotlib.pyplot as plt

months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']
sales = [12000, 15000, 18000, 14000, 20000, 22000]

plt.plot(months, sales, marker='o')
plt.xlabel("Month")
plt.ylabel("Sales")
plt.title("Monthly Sales - Line Plot")
plt.show()
```

OUTPUT

IDLE Shell 3.12.3

File Edit Shell Debug Options Window Help

```
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/Monthlysales.py
```

2.Bar Plot of Monthly Sales

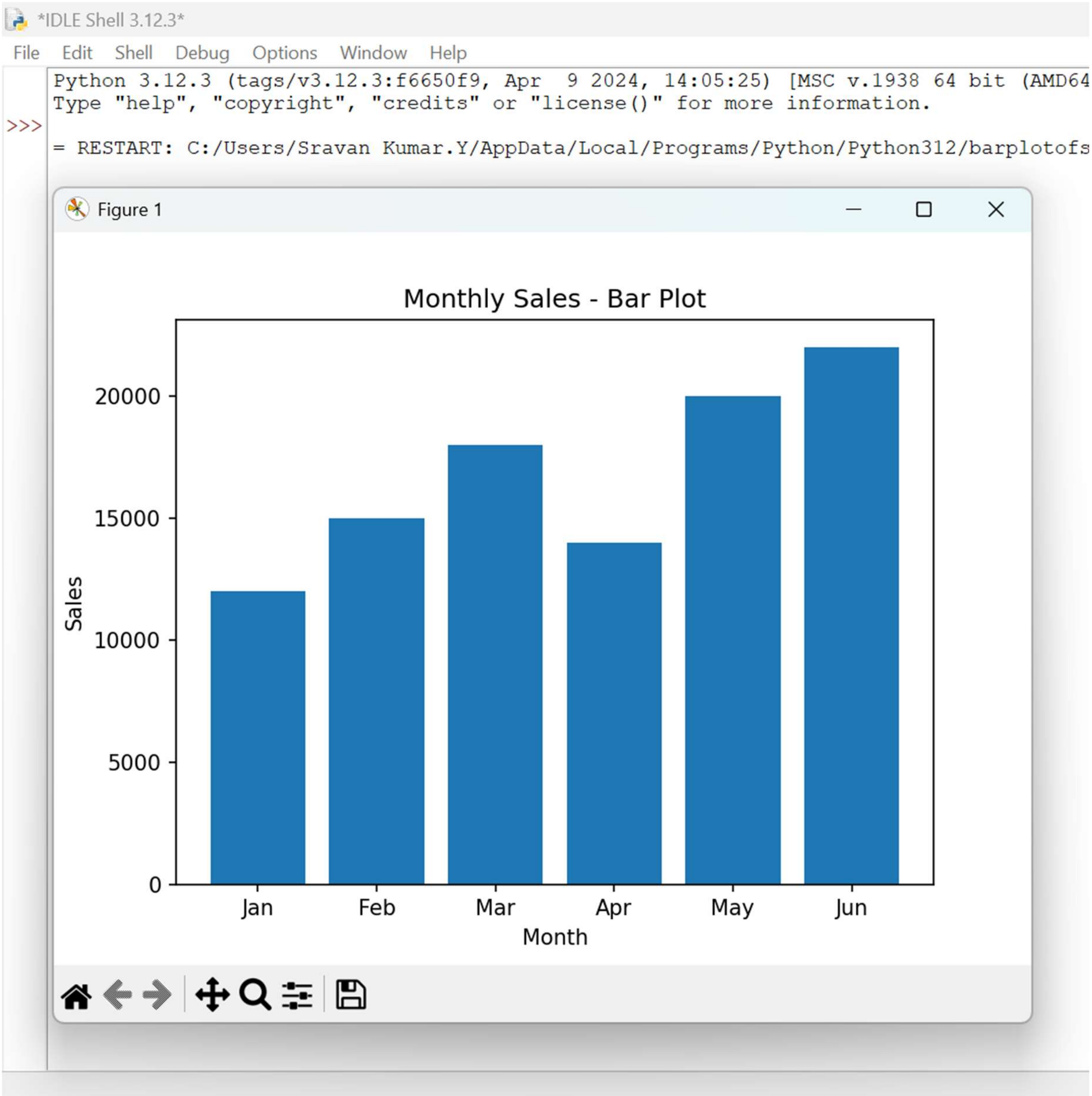
CODE

barplotofsales.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/barplotofsales.py (3.12.3)

File Edit Format Run Options Window Help

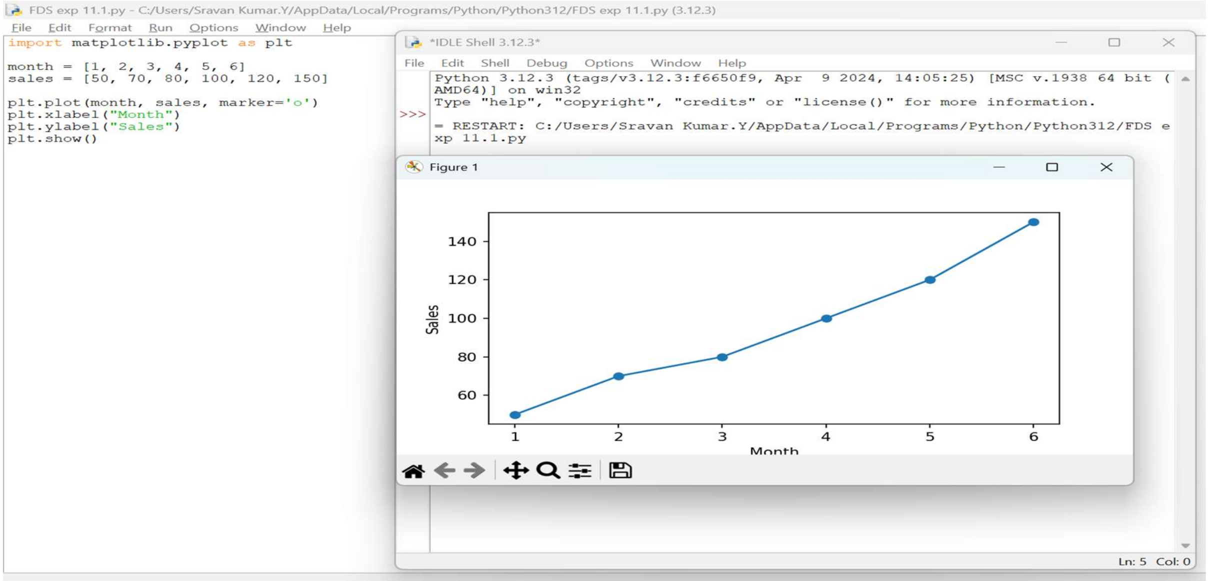
```python
import matplotlib.pyplot as plt

months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']
sales = [12000, 15000, 18000, 14000, 20000, 22000]

plt.bar(months, sales)
plt.xlabel("Month")
plt.ylabel("Sales")
plt.title("Monthly Sales - Bar Plot")
plt.show()
```

OUTPUT

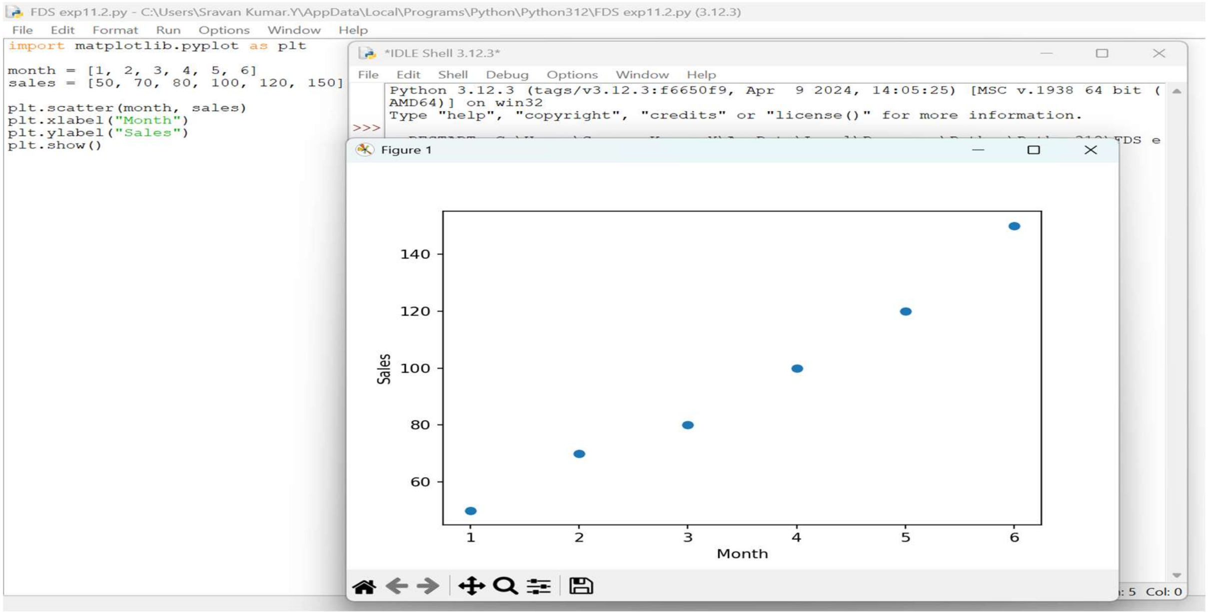


Experiment 11 Sales Visualization using Matplotlib

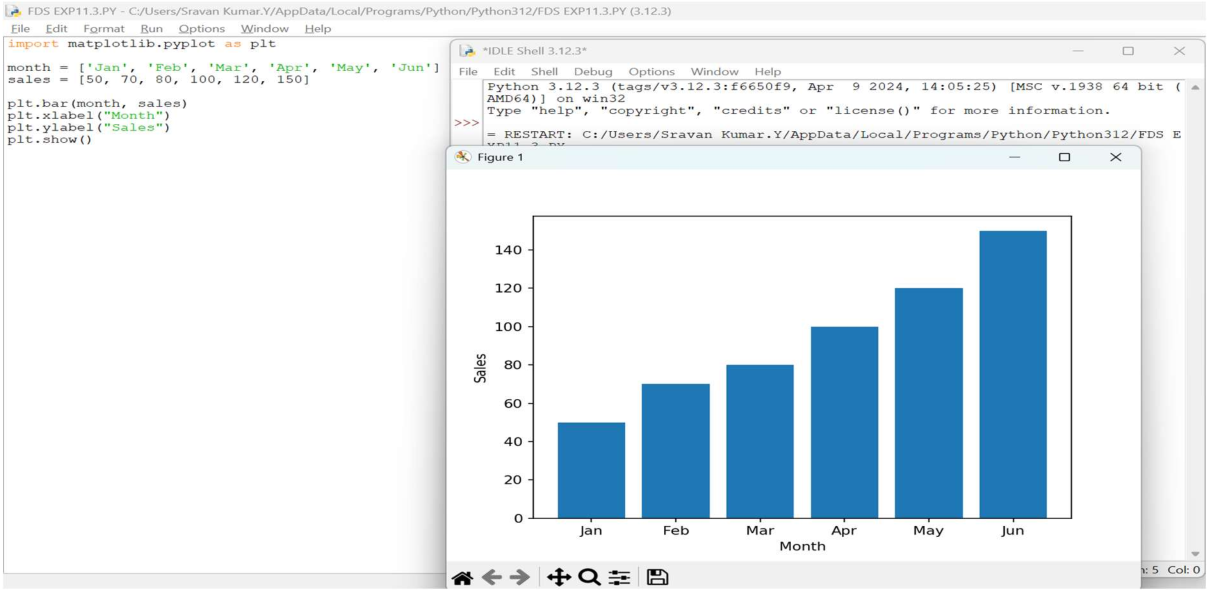
1. Simple Line Plot



2. Scatter Plot



3.Bar Plot



Experiment 12

1.Line Plot of Monthly Temperature

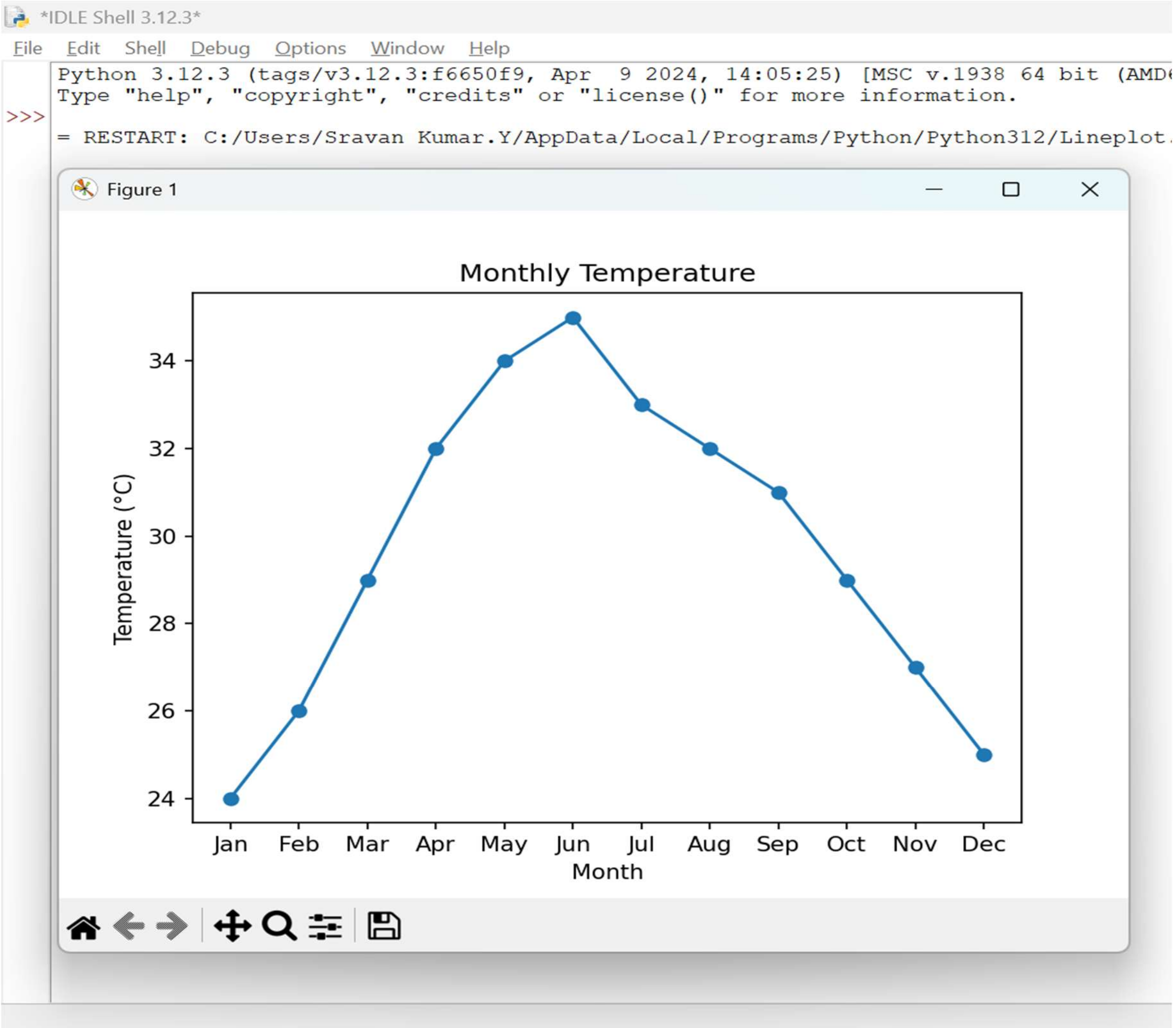
CODE

Lineplot.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/Lineplot.py (3.12.3)

File Edit Format Run Options Window Help

```
import matplotlib.pyplot as plt

months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun',
          'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec']

temperature = [24, 26, 29, 32, 34, 35, 33, 32, 31, 29, 27, 25]

plt.plot(months, temperature, marker='o')
plt.xlabel("Month")
plt.ylabel("Temperature (°C)")
plt.title("Monthly Temperature")
plt.show()
```

OUTPUT



2. Scatter Plot of Monthly Rainfall

FDS exp 12.2.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDS exp 12.2.py (3.12.3)

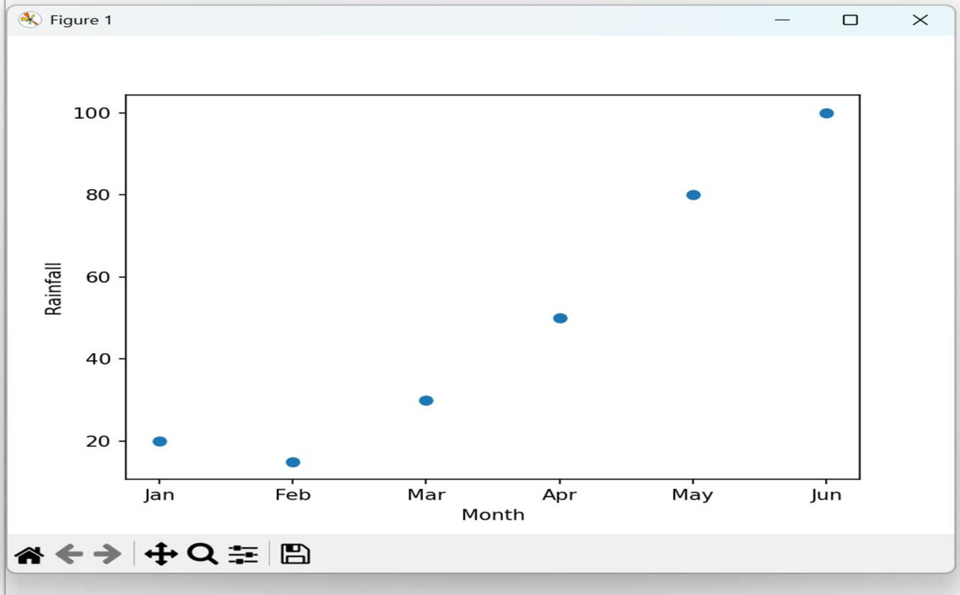
File Edit Format Run Options Window Help

```
import matplotlib.pyplot as plt

months = ['Jan','Feb','Mar','Apr','May','Jun']
rainfall = [20,15,30,50,80,100]

plt.scatter(months, rainfall)
plt.xlabel("Month")
plt.ylabel("Rainfall")
plt.show()
```

IDLE Shell 3.12.3

File Edit Shell Debug Options Window Help

```
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDS exp 12.2.py
```



13. Word Frequency Distribution

CODE

WordfrequencyDis.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDS Exp 13.py/WordfrequencyDis.py (3.12.3)

File Edit Format Run Options Window Help

```python
from collections import Counter

file = open("sample_text.txt", "r")
text = file.read().lower()
file.close()

words = text.split()
frequency = Counter(words)

for word, count in frequency.items():
    print(word, ":", count)
```

IDLE Shell 3.12.3

File Edit Shell Debug Options Window Help

```
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDS E
xp 13.py/WordfrequencyDis.py
python : 2
is : 2
easy. : 1
powerful. : 1
>>>
```

14.Frequency Distribution of Customer Ages

FDS Exp 14.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDS Exp 14.py (3.12.3)

File Edit Format Run Options Window Help

```python
import pandas as pd

data = {'Age': [20, 25, 20, 30, 25, 20, 35]}
df = pd.DataFrame(data)

print(df['Age'].value_counts().sort_index())
```

IDLE Shell 3.12.3

File Edit Shell Debug Options Window Help

```
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDS E
xp 14.py
Age
20    3
25    2
30    1
35    1
Name: count, dtype: int64
>>>
```

15. Frequency Distribution of Likes

FDS exp 15.py - C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDS exp 15.py (3.12.3)

File Edit Format Run Options Window Help

```python
import pandas as pd

data = {'Likes': [10, 20, 10, 30, 20, 10, 40]}
df = pd.DataFrame(data)

print(df['Likes'].value_counts().sort_index())
```

IDLE Shell 3.12.3

ile Edit Shell Debug Options Window Help

```
Python 3.12.3 (tags/v3.12.3:f6650f9, Apr  9 2024, 14:05:25) [MSC v.1938 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>
= RESTART: C:/Users/Sravan Kumar.Y/AppData/Local/Programs/Python/Python312/FDS e
xp 15.py
Likes
10    3
20    2
30    1
40    1
Name: count, dtype: int64
>>
```